



FINAL REPORT

Building Semantic Corpus of Air Quality Research Literature in India using Semantic Climate Toolkits

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I. ABSTRACT

The rapid expansion of scientific literature has created a pressing need for efficient, scalable, and reproducible methods to organize and analyze research data. Semantic corpora provide a structured solution by integrating systematic literature retrieval with metadata organization, enabling researchers to manage large volumes of publications and support data-driven analysis. This study presents the development of a semantic corpus focused on Air Quality Index (AQI) research in the Indian context, conducted as part of the semanticClimate internship.

The methodology involved designing a Boolean research query targeting ambient air pollution, AQI, and particulate matter (PM_{2.5} and PM₁₀) within India. The query, named *aqi_india_pilot*, was executed using the *semantic_corpus* toolkit (*pygetpapers*) on the Europe PMC repository, with a result limit of 50 publications and a requirement for full-text XML availability. The workflow included automated literature retrieval, metadata extraction, relevance scoring, and XML verification. A total of 50 publications were retrieved, of which 18 achieved a relevance score of five or higher, and 47 had XML full-text availability.

Following automated processing, a manual screening phase was conducted to ensure contextual relevance. Publications were evaluated based on titles, abstracts, and metadata, and were annotated as “include” or “exclude.” As a result, 20 papers were selected for inclusion in the final corpus. The selected literature was organized into structured review tables in CSV and JSON formats, capturing key attributes such as metadata, relevance scores, XML availability, and review status. Additionally, an HTML-based review viewer was developed to enhance accessibility and enable efficient navigation of the dataset.

A key feature of this work is its emphasis on reproducibility. The parameterized query design and version-controlled outputs allow the workflow to be easily re-executed, refined, or extended. This enables comparison across multiple query iterations using publication identifiers such as PMCID and PMID, facilitating tracking of newly added or overlapping studies. The final corpus, along with its metadata and documentation, was stored in a GitHub repository to support collaboration and reuse.

The resulting semantic corpus provides a structured and extensible foundation for advanced computational applications, including BagIt packaging, text chunking, embedding generation, and Retrieval-Augmented Generation (RAG)-based systems for AQI research. Overall, this work demonstrates an effective approach to large-scale literature management, combining automation, structured data organization, and manual validation to support future research and AI-driven knowledge discovery.

Keywords: Semantic corpus, Air Quality Index, *pygetpapers*, Europe PMC, PM_{2.5}, PM₁₀, reproducible research, metadata, RAG.

II. INTRODUCTION

The semanticClimate project focuses on building structured, machine-readable, and semantically enriched corpora from large-scale scientific literature related to climate and environmental domains. Its primary goal is to enable efficient organization, retrieval, and analysis of research data by transforming unstructured documents into structured knowledge systems. By integrating techniques such as semantic annotation, metadata generation, and knowledge representation, the project supports advanced research workflows and data-driven decision-making across a range of environmental and public-health topics.

Within the domain of air quality and environmental research, a vast number of scientific papers contain critical information about the Air Quality Index (AQI), particulate matter (PM_{2.5}, PM₁₀), pollution sources, and their impact on human health and the environment. However, this information is typically distributed across multiple documents in unstructured formats, making it difficult to access, analyse, and utilise effectively for computational purposes. Air pollution in India, in particular, is a subject of intense and growing scientific interest, spanning atmospheric science, public health, urban planning, and environmental policy, and the corresponding literature is correspondingly large and fragmented.

To address this challenge, the present work focuses on constructing a semantic corpus from research papers using the semanticClimate semantic_corpus toolkit (pygetpapers). The methodology involves retrieving relevant papers through query-based filtering of Europe PMC, screening them for relevance and full-text (XML) availability, and organising them into a structured review table in preparation for BagIt corpus construction. This structured corpus lays the foundation for future work, including the development of a Retrieval-Augmented Generation (RAG) based chatbot system, which will be taken up in the next phase of the project.

Semantic Corpora

A corpus is a structured collection of documents assembled for a specific purpose. In computational research, corpora often consist of text documents accompanied by metadata and annotations. A semantic corpus extends this concept by incorporating meaningful relationships between documents, entities, concepts, and metadata, allowing richer exploration and analysis than a simple document repository.

Semantic corpora provide several advantages:

- Standardised organisation of scientific literature.
- Improved retrieval through metadata and semantic relationships.
- Support for reproducible research workflows.
- Compatibility with natural language processing and machine learning techniques, including Retrieval-Augmented Generation.
- Facilitation of systematic reviews and evidence synthesis.
- Integration with knowledge graphs and semantic web technologies.

Air Quality Index and Particulate Matter Literature

The Air Quality Index is a composite indicator used to communicate the level of air pollution and its associated health risks to the public. It is derived from concentrations of key pollutants, most notably PM_{2.5} and PM₁₀, along with gases such as NO₂, SO₂, CO, and O₃. Scientific literature on AQI spans predictive modelling (including machine learning and physics-informed neural approaches), epidemiological studies linking pollutant exposure to respiratory and cardiovascular outcomes, structural health monitoring in polluted urban environments, and policy-oriented assessments of exposure bias in city-level reporting. Because this literature is scattered across many journals and repositories, systematic retrieval and organisation is essential before it can be used for downstream applications such as knowledge graphs or conversational retrieval systems.

Reproducible Research and FAIR Principles

Scientific reproducibility is a fundamental principle of modern research. Reproducible workflows allow independent researchers to obtain comparable results by following the same methodology using the same data and computational procedures. In literature-based research, reproducibility requires clear documentation of search strategies, repository sources, screening criteria, and data organisation methods.

The FAIR Data Principles — Findable, Accessible, Interoperable, and Reusable — provide internationally recognised guidelines for research data management. Applying these principles to semantic corpus construction improves transparency, facilitates collaboration, and enhances the long-term value of research datasets. Within this internship, reproducibility was emphasised through a named, version-tracked query (`aqi_india_pilot`), a fixed result limit and required full-text format, and a documented, revisable screening workflow.

Systematic Literature Retrieval and Screening

Systematic approaches to literature retrieval, of the kind formalised in evidence-synthesis frameworks such as PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses), emphasise a transparent chain from search strategy to final included set: a query is defined and logged, records are retrieved from one or more databases, duplicates and clearly irrelevant records are removed, remaining records are screened against explicit inclusion and exclusion criteria, and the number of records at each stage is reported. The workflow adopted in this project mirrors this logic at a smaller scale — a single named query against Europe PMC, automated relevance scoring, an XML-availability check, and a manual include/exclude decision — while remaining lightweight enough to be re-run and refined many times over the course of an internship.

This systematic framing matters because the ultimate purpose of the corpus is not simply to store papers, but to support a future Retrieval-Augmented Generation (RAG) chatbot. A RAG system is only as reliable as the corpus it retrieves from: papers that are off-topic, duplicated, or available only as abstracts (rather than full text) can degrade the quality of retrieved context and, in turn, the quality of generated answers. Building relevance scoring and full-text availability checks into the retrieval stage itself, rather than leaving them to be discovered later, was therefore treated as a priority from the outset of this work.

Why Europe PMC

Europe PMC was selected as the primary repository for this pilot for three main reasons. First, its coverage of biomedical, life-science, and increasingly environmental-health literature is extensive and includes a large proportion of Indian-authored public-health and atmospheric-science research relevant to AQI. Second, it exposes a well-documented REST API that the semantic_corpus toolkit can query programmatically, returning structured metadata (title, authors, journal, publication year, DOI, PMCID/PMID) alongside indicators of full-text availability. Third, a meaningful subset of its records provide full-text XML rather than only an abstract, which is a prerequisite for the text-chunking and embedding steps planned for the next phase of the project. Therefore, Europe PMC was considered the most suitable repository for this study because of its strong coverage of biomedical and public-health literature, structured metadata, and availability of full-text XML records required for subsequent corpus processing and analysis.

III. OBJECTIVES

The internship was designed around the following objectives:

1. To design and implement a structured Boolean query for retrieving AQI-related research publications from the Europe PMC repository.
2. To utilize the semantic_corpus toolkit (pygetpapers) for automated literature retrieval along with comprehensive metadata collection.
3. To extract and organize key bibliographic metadata, including title, authors, year, journal, abstract, DOI, and repository identifiers.
4. To apply automated relevance scoring and verify full-text XML availability to prioritize high-quality and machine-readable publications.
5. To conduct systematic manual screening of retrieved papers and classify them as “include” or “exclude” based on relevance.
6. To generate structured review tables in CSV and JSON formats capturing metadata, relevance scores, XML availability, and review status.
7. To develop an HTML-based review viewer for improved accessibility, visualization, and navigation of the literature dataset.
8. To establish a reproducible and version-controlled workflow using parameterized queries and GitHub-based storage.
9. To enable iterative query refinement and cross-run comparison using identifiers such as PMCID and PMID.
10. To prepare the semantic corpus for downstream applications including BagIt packaging, text processing, and Retrieval-Augmented Generation (RAG)-based systems.

IV. DETAILED OVERVIEW OF WORK:

4.1. Flow chart:

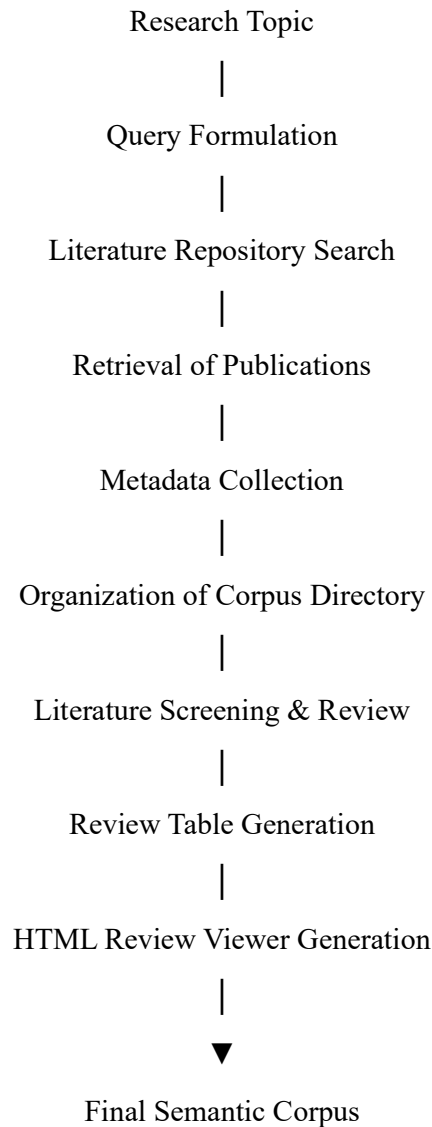


Figure 4.1. Overview of the project workflow

4.2 Environment Setup

The `semantic_corpus` toolkit (pygetpapers), developed by semanticClimate, was installed directly from its GitHub repository (https://github.com/semanticClimate/semantic_corpus), along with supporting libraries such as `pandas` and `ipywidgets`, in a Google Colab environment. Google Colab was chosen for its zero-configuration setup, free access to compute, and ease of sharing notebooks with the semanticClimate team for review. The toolkit itself is a Python 3.8+ package built on `requests`,

beautifulsoup4, lxml, tqdm, configparser, and pyyaml, and exposes both a command-line interface and a Python API for corpus creation, searching, and downloading.

At a package level, semantic_corpus organises its functionality into four components: a core module handling corpus management and repository interfaces, a repositories module implementing the Europe PMC and arXiv connectors, a storage module handling BagIt packaging, and a command-line interface (cli.py) that exposes the create, search, and download subcommands used throughout this project.

Working inside Google Colab rather than a local machine had several practical benefits specific to this internship. It removed the need to manage Python versions and virtual environments on a personal laptop; it allowed the notebook, intermediate CSV/JSON outputs, and generated review tables to be shared directly with the supervising team for feedback; and it provided enough memory and network bandwidth to comfortably retrieve fifty full-text-checked records from Europe PMC in a single run. The trade-off was that the environment is ephemeral by default, so all meaningful outputs — the review table, the query configuration, and the downloaded metadata — were exported out of the notebook and pushed to the project's GitHub repository at the end of each session, rather than left to depend on the Colab runtime persisting.

4.3 Selection and Definition of the Research Query

The first step in the corpus development process involved selecting an appropriate and well-defined research query that would effectively capture relevant literature while maintaining specificity. Air pollution in the Indian context was chosen as the focal topic due to its high scientific and societal significance, as well as its interdisciplinary nature. The issue of air quality intersects multiple domains, including atmospheric science, environmental science, public health, urban planning, and climate studies, making it a rich and impactful area for research. Additionally, the increasing volume of peer-reviewed publications on air pollution and AQI-related studies provided a strong foundation for building a meaningful and representative corpus.

To systematically retrieve relevant publications, a Boolean query was carefully constructed to target key concepts associated with air quality research. The query combined terms related to ambient air pollution, Air Quality Index (AQI), particulate matter (PM_{2.5} and PM₁₀), and contextual environmental factors such as climate, meteorology, and seasonal variations, all within the geographic scope of India:

("ambient air pollution" OR AQI) AND (India) AND (PM_{2.5} OR PM₁₀) AND (climate OR environment OR seasonal OR meteorology)

This query was implemented programmatically using the semantic_corpus workflow, where it was parameterized through a structured configuration. A unique query name (*aqi_india_pilot*) was assigned, along with a result limit of 50 publications. The Europe PMC repository was selected as the data source, and a constraint was applied to retrieve only those publications available in XML format to support downstream processing. Additional parameters included defining the output directory path for storing results and allowing scope for future refinement through a revision parameter. The configuration ensured that all query outputs were systematically stored in a dedicated directory, enabling organized data management and easy access to retrieved records.

By defining the query in a parameterized and programmatic manner—rather than relying on ad-hoc manual searches—the workflow ensures that every aspect of the retrieval process is transparent and reproducible. This approach allows future researchers or collaborators to re-run the same query under identical conditions, modify parameters for refined searches, and systematically compare results across different iterations. Furthermore, storing the query configuration and outputs within a structured directory supports version control and collaborative development. Overall, this step laid a robust and scalable foundation for the subsequent stages of literature retrieval, screening, and semantic corpus construction.

```
In [ ]: from pathlib import Path

QUERY_NAME = "aqi_india_pilot"
QUERY_STRING = (
    '("ambient air pollution" OR AQI) '
    'AND (India) '
    'AND (PM2.5 OR PM10) '
    'AND (climate OR environment OR seasonal OR meteorology)'
)
LIMIT = 50
FORMATS = ["xml"]
REPOSITORY = "europe_pmc"
NOTES = ""
REVISION_OF = None # e.g. "aqi_india_pilot" when refining a prior run

OUTPUT_DIR = Path("/content/queries", QUERY_NAME)
OUTPUT_DIR

Out[ ]: PosixPath('/content/queries/aqi_india_pilot')
```

Figure 4.2 Overview of the pipeline with query

4.4 Literature Retrieval

After defining the research query, relevant scientific publications were retrieved from academic literature repositories using the semantic corpus workflow.

The retrieval process aimed to identify scientific publications relevant to air quality, Air Quality Index (AQI), and particulate matter (PM2.5 and PM10) in the Indian context while preserving the publication metadata required for subsequent stages of corpus construction.

The retrieved information included:

- Article title
- Authors
- Publication year
- Journal

- Abstract
- DOI
- Repository information
- Keywords
- Publication identifiers

Maintaining complete metadata ensured that publications could be efficiently organized and reviewed later in the workflow.

4.5 Corpus Directory Organization

A standardized directory structure was created to organize all resources associated with the corpus. Maintaining a consistent folder hierarchy improves reproducibility and simplifies navigation.

The corpus is created at [semantic_corpus/corpora/subhangi_project at subhangi · semanticClimate/semantic_corpus](#)

4.6 Metadata Collection

Metadata collection represented one of the most important stages of corpus construction.

Rather than storing only PDF files, each publication was accompanied by structured metadata describing its bibliographic information.

The metadata fields included:

Table 4.1. List of the metadata fields and descriptions

Field	Description
Title	Research article title
Authors	List of authors
Year	Year of publication
Journal	Publishing journal
DOI	Digital Object Identifier
Abstract	Article summary
Keywords	Indexing terms
Repository	Source repository

Organizing publications using metadata provides several advantages:

- Efficient searching
- Filtering by publication year
- Sorting by author
- Grouping publications

- Integration with computational tools
- Future knowledge graph construction

Metadata files were maintained in structured formats that could be easily processed by automated scripts.

The metadata is available on GitHub at [semantic_corpus/corpora/subhangi_project](https://github.com/semanticClimate/semantic_corpus/corpora/subhangi_project) at [subhangi · semanticClimate/semantic_corpus](https://github.com/semanticClimate/semantic_corpus)

4.7 Literature Screening

Following literature retrieval, publications underwent a screening process to determine their relevance to the selected research topic.

Screening involved examining publication titles, abstracts, and associated metadata.

The objective of screening was to remove publications that were unrelated or only marginally relevant to the research question while retaining studies directly addressing air quality index and air pollution in India.

The screening workflow consisted of:

1. Initial retrieval of publications.
2. Duplicate identification.
3. Title screening.
4. Abstract screening.
5. Eligibility assessment.
6. Selection for review.

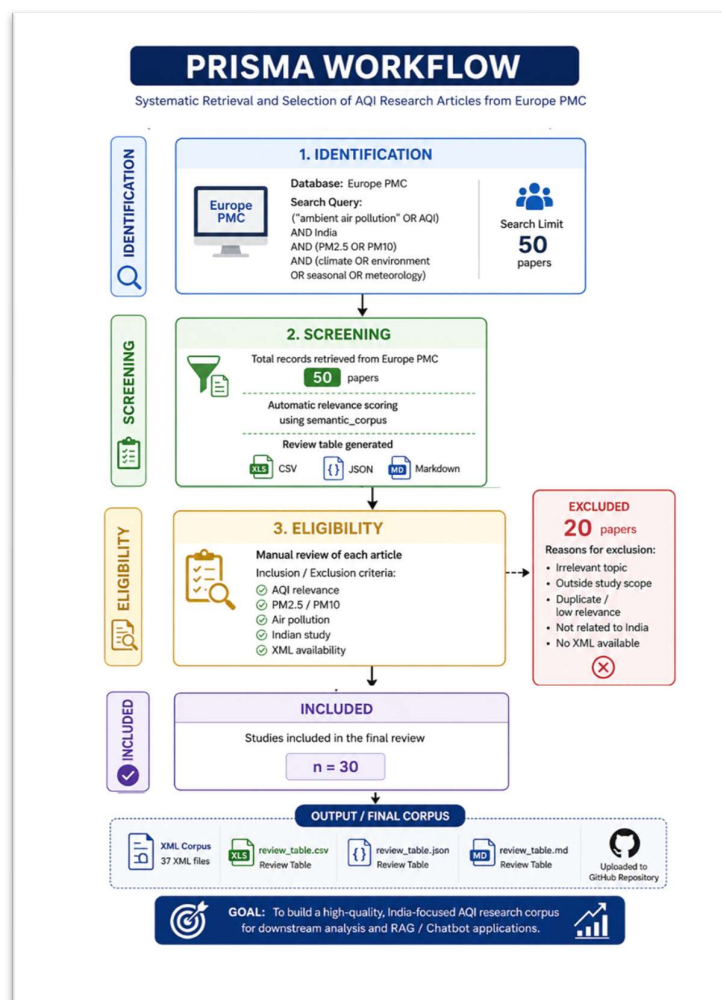


Figure 4.3. Simplified literature screening workflow using PRISMA

Link : [semantic_corpus/corpora/subhangi_project/prisma_aqi.png at subhangi · semanticClimate/semantic_corpus](#)

4.8 Review Table Generation

To facilitate efficient evaluation of retrieved publications, review tables were generated during the corpus construction process.

The review table served as a structured summary of all publications included within the corpus.

Each record typically contained:

- Article title
- Authors
- Publication year
- Journal
- Repository
- Review status
- Relevance
- Notes

The review table enabled systematic comparison between publications and simplified literature assessment.

An HTML-based review interface was subsequently generated from these review tables to improve accessibility and navigation.

The review_table is available at [semantic_corpus/corpora/subhangi_project/review_table \(2\).csv at subhangi · semanticClimate/semantic_corpus](#)

Table 4.2. Review table for sample papers

review_status	score	pmcid	doi	title	has_xml
include	9	PMC12448019	10.1016/j.mex.2025.103597	A hybrid physics-informed neural and explainable AI approach for scalable and interpretable AQI predictions.	TRUE
include	8	PMC12098713	10.1038/s41598-025-00814-9	Assessment and forecasting of particulate matter emissions and structural health monitoring of	TRUE

				buildings in Bangalore.	
include	7	PMC11883379	10.1038/s41598-025-26663-0	Assessing Nexus between Human Health and Air Pollutants through Spatiotemporal Analysis of Particulate Matters.	TRUE
include	6	PMC12995173	10.4103/lungindia.lungindia_509_25	A cross-sectional study on the respiratory health status and its determinants among rural handloom workers in Tamil Nadu.	TRUE
include	6	PMC13199542	10.1038/s41598-026-40057-w	Systematic exposure bias in city-level air quality reporting revealed by 38 years of Indian observations.	TRUE

4.9 HTML Review Viewer

One of the notable components of the workflow was the generation of an HTML-based review viewer. This interactive interface allowed users to browse and inspect the review table through a web browser.

The HTML viewer improved usability by presenting publication metadata in an organized format and supporting easier navigation compared with raw JSON files.

Features of the review viewer included:

- Structured tabular presentation.
- Easy navigation between records.
- Display of publication metadata.
- Review status indicators.
- Improved readability for manual assessment.

Available on GitHub as [semantic_corpus/corpora/subhangi_project/review_table_html/index.html](https://github.com/subhangi/project/review_table_html/index.html) at [subhangi · semanticClimate/semantic_corpus](#)

Status	Title	PMCID	Authors	Date	XML	PDF	Read	Abstract
include	Sustainable modular biofiltration system with rainshower technology for AQI reform.	PMC12086192	Jaiswal V; Singh S; Mangal A; Singh PP	2025-05-18	yes	no	HTML Read	India is the tenth most polluted nation in the world, according to the World Health Organization (WHO) 2022 assessment. Approximately 1.67 million deaths are caused by lung, cardiovascular, stroke, an...
include	A hybrid physics-informed neural and explainable AI approach for scalable and interpretable AQI predictions.	PMC12448019	Chandrashekar SV; Khan F; Sridhar S; Dhanaraj RK; Balusamy B; Pamucar D; Sathyamoorthy M	2025-08-29	yes	no	HTML Read	<i>Air Pollution is a critical environmental issue affecting public health, climate, and ecosystems. However, accurately predicting and classifying Air Quality Index (AQI) levels across different regi...
include	Assessment and forecasting of particulate matter emissions and structural health monitoring of buildings in Bangalore.	PMC12098713	Devi LP; Chandana R; Bandhu D	2025-05-22	yes	no	HTML Read	Particulate Matter (PM) emissions have emerged as a critical global concern due to rapid urbanisation, increased vehicular traffic, and construction activities. These emissions not only harm human hea...
include	Impact of Housing Conditions and Air Quality Awareness with Indoor PM2.5 Levels in Urban Jodhpur.	PMC12815361	Huda RK; Gupta R; Kumar P; Singh J; Sharma AK; Toteja GS; Babu BV	2025-11-13	yes	no	HTML Read	Indoor air pollution poses a major public health concern in India, particularly among urban populations exposed to high levels of particulate matter. This study aimed to assess the influence of housin...

Figure 4.4. HTML review table generated during corpus construction.

4.10 Semantic Corpus Organisation

After screening and review, the selected publications were integrated into the final semantic corpus.

The completed corpus consisted of:

- Structured directory hierarchy.
- Downloaded literature.
- Metadata files.
- Review tables.
- HTML review interface.
- Supporting documentation.

The organization ensured that the corpus could be reused by other researchers without requiring reconstruction of the workflow.

V. OT-VIRARe TG and the Chilean National Committee

I actively participated in the demonstration of the chatbot representing the Climate Insight and the semantic corpus. This meeting called together the discussions on the IPCC chatbot and the semantic corpus with CODATA and other academicians across Argentina and Chile. Throughout the event, I contributed as a participant and enhanced my technical understanding of open-access data processing and global research ecosystem.

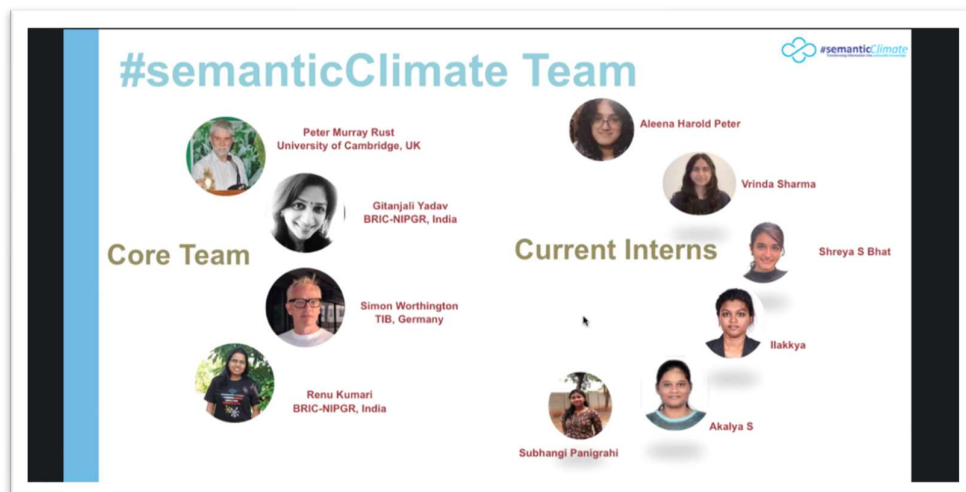


Figure 5.1. Current team of semanticClimate

VI. RESULTS

The reviewed papers were stored as XML files in the GitHub corpus for the development of the encyclopedia and chatbot. This can further be used by other scientists and enthusiasts for continued progress of semanticClimate.

The query `aqi_india_pilot` was run against Europe PMC with a result limit of 50 papers and a required full-text format of XML. Table 6.1 summarises the outcome of this run.

Table 6.1. Query run summary

Metric	Value
Query name	<code>aqi_india_pilot</code>
Repository	Europe PMC
Result limit	50
Required format	XML
Total papers retrieved (row_count)	50

Papers with score ≥ 5 (high_score_count)	18
Papers with XML available (xml_count)	47
Papers with score ≥ 5 AND XML available	17
Papers manually marked "include"	20

Research Areas and Relevance of Selected Literature

The literature selected for further analysis covers several interconnected research areas within air-quality research in India, including Air Quality Index (AQI) prediction, particulate matter assessment, environmental monitoring, seasonal variation, and public-health impacts of air pollution. Chandrashekar SV, Khan F, Sridhar S, Dhanaraj RK, Balusamy B, Pamucar D, and Sathyamoorthy M (2025) investigated AQI prediction and classification using a hybrid physics-informed and explainable artificial intelligence approach, making the study relevant to the AQI prediction component of the query. Devi LP, Chandana R, and Bandhu D (2025) examined particulate matter emissions and AQI forecasting in Bangalore, including PM_{2.5} and PM₁₀ and their seasonal variations, which directly relates to the particulate-matter, India, and seasonal aspects of the query. Vinoth Kumar V, Chitra A, and Costantia Dovita S (2026) studied respiratory health among rural handloom workers in Tamil Nadu, contributing to the public-health aspect of air-pollution research and its effects on exposed populations. Sharma G, Parmar PD, Nema AK, Gargava P, and Kota SH (2026) analysed long-term city-level air-quality reporting using Indian observations, which is relevant to understanding air-quality monitoring and reporting in the Indian context. Collectively, these selected studies represent important areas related to AQI, particulate matter, environmental monitoring, seasonal variation, and health impacts of air pollution, supporting their relevance to the objective of developing a structured semantic corpus of air-quality research in India.

The copy of the query is available on [semantic_corpus/corpora/subhangi_project at subhangi · semanticClimate/semantic_corpus](#)

On GitHub : [semantic_corpus/corpora/subhangi_project/aqi_india_pilot \(2\)/aqi_india_pilot at subhangi · semanticClimate/semantic_corpus](#)

PM11791089.xml	updated review table html	last week
PM11861530.xml	updated review table html	last week
PM11883379.xml	updated review table html	last week
PM12086192.xml	updated review table html	last week
PM12098713.xml	updated review table html	last week
PM12141506.xml	updated review table html	last week
PM12251630.xml	updated review table html	last week
PM12328754.xml	updated review table html	last week
PM12329590.xml	updated review table html	last week
PM12355607.xml	updated review table html	last week
PM12364328.xml	updated review table html	last week
PM12368213.xml	updated review table html	last week
PM12448019.xml	updated review table html	last week

Figure 6.1. Reviewed pygetpapers as XML files

However, running the pipeline against the query gives the details of the papers in csv, json, CSS, HTML and md files.

csv : [semantic_corpus/corpora/subhangi_project/review_table\(2\).csv at subhangi · semanticClimate/semantic_corpus](#)

json : [semantic_corpus/corpora/subhangi_project/review_table_html/review_table.json at subhangi · semanticClimate/semantic_corpus](#)

html : [semantic_corpus/corpora/subhangi_project/review_table_html at subhangi · semanticClimate/semantic_corpus](#)

VII. TOOLS AND TECHNOLOGIES USED

Python

During the internship, Python scripts were used to support literature processing, metadata handling, review table generation, and organization of the semantic corpus. Python also enabled automation

Git and GitHub

Git was used to monitor changes in project files, maintain a history of modifications, and facilitate collaborative development. The project repository was managed using GitHub, allowing organized maintenance of scripts, documentation, and corpus files. Version control ensured that updates to the corpus could be tracked systematically and that previous versions of the workflow could be recovered when required.

GitHub Codespaces and Visual Studio Code

The computational workflow was implemented using **GitHub Codespaces**, which provided a cloud-based development environment configured with the required software dependencies.

Visual Studio Code (VS Code) served as the integrated development environment (IDE) for editing Python scripts, managing project files, and executing workflow components. Features such as syntax highlighting, integrated terminal access, and source control integration improved productivity and facilitated efficient project management.

Slack

The slack was used as a platform for communication, progress updates and coordination with the team members

CSV and JSON

CSV and JSON were used as the primary structured data formats for the review table, chosen for their lightweight syntax, broad tool support, and suitability for both spreadsheet-based manual review and downstream programmatic processing.

HTML-Based Review Viewer

An HTML review viewer was generated to improve the accessibility of the corpus. Rather than examining raw JSON files, users could inspect publication metadata through a structured web interface. The HTML review viewer demonstrated how computational visualization can enhance literature management without requiring specialized software.

Literature Repositories (Europe PMC)

Scientific publications included in the corpus were obtained through established literature repositories integrated into the workflow. These repositories provide access to peer-reviewed research articles and associated bibliographic metadata.

Markdown Documentation

Markdown files were used to document the workflow, project structure, and corpus organization. Their lightweight syntax facilitated clear documentation while remaining compatible with GitHub and other version control platforms.

pandas and ipywidgets

pandas was used to load, filter, and inspect the review table as a DataFrame, enabling quick computation of summary statistics such as `high_score_count` and `XML_count`. ipywidgets supported interactive inspection of results within the Google Colab notebook environment.

Google Colab

Google Colab provided a cloud-based, zero-configuration environment for running the `semantic_corpus` toolkit and its dependencies, and facilitated easy sharing of the notebook with the semanticClimate team for review and feedback.

VIII. CONCLUSION

The project successfully established a reproducible and structured pipeline for querying, scoring, and reviewing AQI-related scientific literature from Europe PMC using the *semanticClimate* *semantic_corpus* toolkit. By implementing a named and version-tracked query with a clearly defined result limit and a requirement for full-text XML availability, each retrieval run could be systematically logged, revisited, and refined without any loss of context. The integration of automated relevance scoring and XML availability checks, combined with careful manual review, enabled the creation of a structured and versioned review table that identified papers suitable for corpus inclusion. This approach ensured a balance between efficiency and accuracy by leveraging automated filtering while retaining the reliability of expert validation.

The workflow's support for iterative query refinement and cross-run comparison further enhances the reproducibility and precision of the literature search process. By tracking new, excluded, and overlapping papers across multiple runs, it becomes possible to quantitatively evaluate how changes in the query strategy affect retrieval outcomes. This not only improves transparency but also enables a more data-driven approach to optimizing search strategies. As a result, the workflow provides a strong and scalable foundation for building a well-curated semantic corpus for air quality research, while also laying the groundwork for the development of a future retrieval-augmented, chatbot-based system for accessing AQI-related knowledge.

From a broader perspective, this internship provided valuable hands-on experience in applying computational and data-driven tools—such as Python, Europe PMC's API, pandas for data inspection and analysis, and Git/GitHub for version control—to a real-world research challenge: managing the rapidly growing and fragmented body of air-quality literature relevant to India. The experience not only strengthened my technical and analytical skills but also improved my understanding of reproducible research practices and collaborative workflows. Importantly, the methods and skills developed during this project are highly transferable and can be extended to other domains within environmental science and public health, contributing to the larger goals of the *semanticClimate* initiative.

IX. REFERENCES

- Semantic Climate. Semantic Corpus Repository. https://github.com/semanticClimate/semantic_corpus
- Europe PMC. <https://europepmc.org/>
- Google Collaboratory. <https://colab.research.google.com/>
- Python Software Foundation. <https://www.python.org/>
- Pandas Documentation. <https://pandas.pydata.org/>
- Jupyter Notebook Documentation. <https://jupyter.org/>
- Europe PMC REST API Documentation. <https://europepmc.org/RestfulWebService>
- Wilkinson MD et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*. 2016.
- The PRISMA 2020 Statement. *BMJ*. 2021. <https://www.prisma-statement.org/prisma-2020>
- Chandrashekar SV, Khan F, Sridhar S, Dhanaraj RK, Balusamy B, Pamucar D, Sathyamoorthy M. (2025). *A hybrid physics-informed neural and explainable AI approach for scalable and interpretable AQI predictions*.
- Devi LP, Chandana R, Bandhu D. (2025). *Assessment and forecasting of particulate matter emissions and structural health monitoring of buildings in Bangalore*. Scientific Reports. DOI: 10.1038/s41598-025-00814-9.
- Vinoth Kumar V, Chitra A, Costantia Dovita S. (2026). *A cross-sectional study on the respiratory health status and its determinants among rural handloom workers in Tamil Nadu*. Lung India. DOI: 10.4103/lungindia.lungindia_509_25.
- Sharma G, Parmar PD, Nema AK, Gargava P, Kota SH. (2026). *Systematic exposure bias in city-level air quality reporting revealed by 38 years of Indian observations*. Scientific Reports. DOI: 10.1038/s41598-026-40057-w.